Chia-Yuan Chiang

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Objective: Third-year EE undergraduate seeking an embedded firmware / FAE / hardware-software integration internship. Hands-on across the full stack — mechanism, CNC machining, PCB, embedded firmware, and data analysis. Available for full-time internship during summer and part-time during the semester.

Education

Yuan Ze University — Taoyuan, Taiwan
B.S. in Electrical Engineering — *Sep. 2024 – Sep. 2027 (Expected)*

Nangang Senior High School — Taipei, Taiwan
Electronics Technology (Vocational, Mechatronics) — *Sep. 2020 – Jun. 2023*

Projects

Open-Source QMK x STM32 Numpad | *C, ChibiOS HAL, EasyEDA* — 2026

- Designed a 17-key PCB in EasyEDA Pro, hand-soldered the board, and flashed STM32F072 via DFU; reverse-engineered ChibiOS HAL config (chconf/halconf/mcuconf) to resolve build conflicts and timing issues between the open-source PCB and QMK firmware.
- Reached stable USB HID enumeration with ¡5ms key latency, working RGB Matrix, and live VIA/VIAL remapping without recompiling.

ICU VAP Early-Warning Model | *PyTorch, LSTM, SHAP* — 2025 – Present

- Single-center pilot (n=109); designed stay-level 5-fold stratified CV that exposed patient-level data leakage in a public benchmark (AUROC dropped from an inflated 0.99 to a real 0.58).
- Used Integrated Gradients to reduce the model to 4 non-invasive features; final LSTM reached 0.98–0.99 AUROC across a 6–72h horizon. Manuscript submitted to IEEE GCCE 2026.

Job Radar | *Python, Playwright, GitHub Actions* — 2026 – Present

- Built a self-driven job aggregator that scrapes 104/Cake/LinkedIn (˜1,300 listings/day) on a daily cron, scores roles against my criteria, and publishes a static dashboard — no backend, runs even with my laptop off.

Swerve Drive Chassis (FRC) | *LabVIEW, CNC, PID Control* — 2020 – 2023

- Designed and machined 3 generations of a swerve-drive chassis from scratch (kinematics, gear train, CNC/wire-EDM fabrication, LabVIEW PID control); won Innovation in Control Award at 2022 FRC Taiwan Hon Hai Regional.

Experience

Undergraduate Researcher — May 2025 – Present
Prof. Shu-Yen Lin's Lab, Yuan Ze University — *Taoyuan, Taiwan*

- Lead the ICU VAP early-warning project (above); responsible for the full pipeline from data cleaning and CV design to model training and feature attribution.

Engineering Advisor & Systems Integrator — 2023 – Present
FRC Team 7645 — *Taipei, Taiwan*

- Own robot system architecture: multi-sensor fusion, PID closed-loop motor control tuning, software/hardware interface definitions; train junior members to debug from root-cause analysis.

Technical Skills

Languages & Firmware: C/C++ (STM32/Cortex-M), Python, QMK/ChibiOS HAL, LabVIEW, Arduino/ESP32, UART/SPI/I2C
EDA / Hardware: Altium Designer, KiCAD, OrCAD, EasyEDA Pro
Machine Learning: PyTorch, TensorFlow/Keras, LSTM/Time-Series, SHAP/XAI, TinyML
Manufacturing: CNC Milling, Mastercam, 3D Printing, Laser Cutting, Welding, AutoCAD, Fusion 360

Awards & Certifications

2022 FRC Taiwan Hon Hai Regional – Finalist Alliance & Innovation in Control Award · 52nd National Skills Competition – Team Project Representative (2022) · 2020 FRC CIT 5G Regional – Excellence in Engineering Award